



Task 3: Real-Time Conversational Avatar Coaching Assistant



Date

Jul 12, 2025



Participants

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This document outlines the full technical implementation and usage instructions for Task 3, a real-time AI coaching assistant built with a conversational avatar. The system features speech input, LLM-powered replies, avatar GIF state transitions, and CRM logging via Airtable.



Tech Stack

Layer	Technology
Frontend	React.js
Backend	Node.js, Express.js, Socket.IO
Speech to Text	Deepgram API
AI Assistant	DeepSeek Chat API
Text to Speech	ElevenLabs API
Avatar	Animated GIFs (Idle, Thinking, Talking)
CRM	Airtable API



Features Overview

- Voice input from user
- Real-time transcription with Deepgram
- AI response generation with DeepSeek
- Audio reply from ElevenLabs
- Female Avatar with expression changes (Idle, Thinking, Talking)
- Automatic conversation logging to Airtable CRM



Setup Instructions

1. Install Dependencies

```
1 npm install
2 cd client
```

```
3 npm install
4 cd ..
```

2. Configure .env

Create a `.env` file in the root with:

```
1 DEEPGRAM_API_KEY=your_deepgram_key
2 DEEPSEEK_API_KEY=your_deepseek_key
3 ELEVENLABS_API_KEY=your_elevenlabs_key
4 AIRTABLE_API_KEY=your_airtable_key
5 AIRTABLE_BASE_ID=your_airtable_base_id
```

3. Run Server (Server.js)

```
1 node server.js
```

4. Run Client (App.js)

```
1 cd client
2 npm start
```

Workflow

```
1 A[🎤 User speaks] --> B[🧠 Deepgram transcribes]
2 B --> C[🗣️ DeepSeek generates reply]
3 C --> D[🗣️ ElevenLabs generates voice]
4 D --> E[👤 Avatar switches to "talking"]
5 C --> F[💬 UI displays text chat]
6 E --> G[🔊 Audio reply sent to UI]
7 C --> H[📁 Log chat to Airtable CRM]
```

Avatar Behavior (GIF Based)

State	Trigger
Idle	Default (no user interaction)
Thinking	After recording is sent to server
Talking	After AI reply + TTS audio is ready

GIFs stored inside `public/avatar/` folder and switched dynamically via React state `avatarState`.

Key Libraries Used

Backend:

- `express`
- `axios`
- `socket.io`

- `dotenv`
- `airtable`

Frontend:

- `react`
- `socket.io-client`

Third-Party Tools

Tool	Purpose
Deepgram	Speech recognition
DeepSeek	Chat LLM (AI brain)
ElevenLabs	Text-to-Speech
Airtable	CRM data storage
Animaker	Avatar GIF creation

Avatar Created with Animaker (Free Tier)

Avatar Type: Female (GIF Style)

Export Format: Animated GIF

Storage Location: `/public/avatar/`

Animaker Pricing (Summary)

Plan	Monthly Price	Features
Free	\$0	5 downloads/month, watermark, limited assets
Starter	\$20/month	10 exports, HD quality, commercial rights, no watermark
Pro	\$49/month	Unlimited exports, 4K, priority support, all features

[More info on Animaker Pricing](#)